Video Transcript

Video 9: Creating a "Hello World" Workflow (07:44)

Let's write a simple "Hello World" workflow. We'll start off by creating a Python file. I'm going to go ahead and create a new file, then choose Python as its type. We'll then go ahead and save it for now into the sample directory, and we'll go ahead and call it abel.py. We'll go ahead and start off with the imports. So, you can see there we're importing Airflow and we are importing the DAG. Then we'll add our operators. Next, we'll do our simple function that we're going to be calling by one of the operators once we write our tasks.

Next, we're going to define our DAG, and you can see there that we're calling it abel. We're starting it two days ago, and our schedule interval is one. Next, we're going to go ahead and add our tasks. You can see there that the very first one is one that instantiates the bash operator, and all we're doing there is we're simply going to write out the date. You can see there as well that the handle to that is going to be t1, and that the task ID that we're giving it is called task one. After that, we're going to have another one that is going to be a bash operator as well, and we're simply going to be sleeping there for five seconds. And then the last one is going to be called task three, the task ID, and you can see there that that one is calling the function that we have here that simply returns five. Then the very last thing to do is that we're going to set the dependencies between the tasks, 1 will run before 2 and 3, and 2 and 3 will run in parallel.

So, let's go ahead and save that file. And next, we need to move that DAG that we just created and direct a cyclic graph into the DAGs directory. The reason we're doing that is because we're sharing that folder, that volume with Airflow, and Airflow is scanning that folder for new DAGs. It does so every 60 seconds, so every minute. So, if you push years into it and you don't see it immediately, don't worry, just give it about a minute or so, and then it should show up. So, I'm going to go ahead and do that now. And as you can see there, I have moved it into the DAGs directory. As you can see here, I have logged in to Airflow, and I am on the landing screen, and you can see there that there is my DAG. Now, at times this can take longer than a minute to show up. But if you haven't seen it in say 10 minutes, you might want to restart your installation. And you can do so by going to your Docker dashboard, and in a minute that will

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show up here, going to the top of it and simply starting it, and starting it again.

Now, this is sort of the really heavyweight type of cycle. But in general, once you give it enough time, it should show up on its own. So, in this case, let's go ahead and select table, and we'll look here at the different ways of looking at the tasks. You can see there that we are on the tree view right now. You can see there that once the DAG starts, we have task one and then task two, and three run in parallel. Now, we can look at it in the graph view as well, and perhaps this one is a more transparent one, and you can zoom there as needed. There's other things that you can do as well. Once you start running, you can look at the calendar view, how long your task duration ran and so on. And you can look at code as well. So, here is our code that we wrote that is now inside of Airflow, inside of that folder that we just took a look at the DAGs folder.

And so now let's go ahead and come back. Let's enable our DAG, give it a chance to run, and we can go ahead and fire it as well manually. I'm going to go ahead and do so. And as you can see there, we get the two days that we're catching up. We started it a couple days back, and then we have triggered it manually as well and so that would be the third run. If we go ahead and take a look at the graph view, we can go ahead, and take a look at the logs of each one of these. And we'll go ahead, and take a look here at the very first log. We're scrolling down and you can see there that the command that ran on the bash was date, and that the output was the following on this system, on this date.

Now, of course that is dependent on what type of system you have. This really could be any time. In fact, one of the problems with machines is that at times you'll start at one odd times, so the date is certainly dependent on that. Now, if we go back to task two, and we take a look at the output of that, we can see here the same and in this case, this simply slept for five seconds, and nothing else was done. And then the very last one, the one that was calling that Python function, if we take a look at it there, you'd see there, this is the wrong one. If we look at task three and we scroll down, you can see there that the function called and all that was returned was that value of five.

So, as you can see here, that gives you a sense here of what the user interface is, how to write a DAG, how to write a workflow and get it into the platform. You can see there that this is simply one. If you started to have many more runs, these are the runs that just ran today, we would be able to get here a much better sense of what's taking place. And if we walked into the workflow

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itself, we'd be able to see a graphical view. If you ever look at the timeline that you see in GitHub repositories of the activity and the errors, perhaps the heat that you might get, you get something similar here and those little squares would be colored by the following things that you see here. So, failed, up for retry, up for reschedule, and many of the other concerns that we typically have.

Now, that we have that we could see here the GitHub type of view that I just mentioned, you can see as well how many the tasks, how long the tasks are taking. Let's see if I can make a little bit smaller view here. You could see the number of tries. In this case, we don't have any of that data, but you could see that once you start to run at scale, this could be pretty useful. This would certainly give you a sense of how your tasks are performing, how your workflows are doing. So, now it's your turn. Go ahead and replicate the code that I have done. Make sure you can get it into your platform. Make sure you can run it and verify by looking at the logs the results.

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